

11-matching-flow: 匹配与网络流 (Matching and Network Flow)

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3 Theorems + 1 Algorithm

To **maximize** the size of a mathematical structure $\mathcal{S}$ in G



Theorem

$\mathcal{S}$ is maximum *iff* G does not contain $\mathcal{S}$ -augmenting objects.

Algorithm

Repeatedly finding $\mathcal{S}$ -augmenting objects until no more ones exist.

To **maximize** the size of a mathematical structure $\mathcal{S}$ in G



To **minimize** the size of its **dual** mathematical structure $\mathcal{S}'$ in G

Theorem (Weak Duality Theorem)

The size of a maximum $\mathcal{S} \leq$ The size of a minimum $\mathcal{S}'$

Theorem (Strong Duality Theorem)

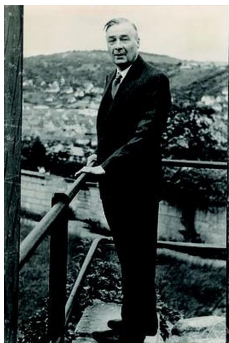
The size of a maximum $\mathcal{S} =$ The size of a minimum $\mathcal{S}'$

let's get
married
today

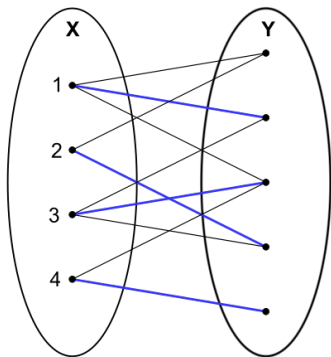


The Marriage Problem (Philip Hall, 1935)

If there is a finite set of **girls**, each of whom knows several **boys**,
under what conditions can all the girls marry boys in such a way that
each girl marries **a boy** that she knows?

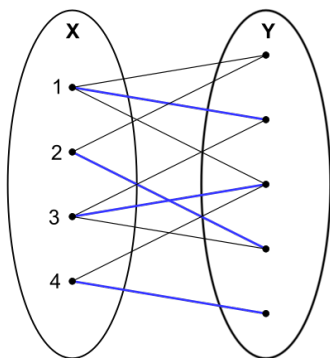


Philip Hall (1904 ~ 1982)



Definition (Matching (匹配))

A **matching** in a graph G is a set of edges with **no shared endpoints**.

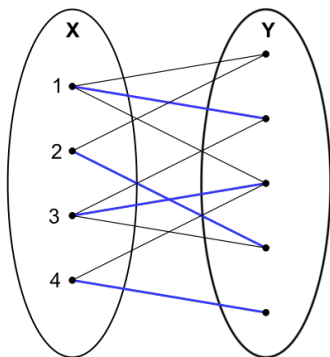


Definition (X -Perfect Matching (X -Saturating Matching))

Let $G = (X, Y, E)$ be a bipartite graph.

An X -perfect matching of G is a matching which covers each vertex in X .

$$|X| \leq |Y|$$



$$\forall W \subseteq X. |W| \leq |N(W)|$$

Theorem (Hall Theorem; 1935)

Let $G = (X, Y, E)$ be a bipartite graph. There is an X -perfect matching of G iff

$$\forall W \subseteq X. |W| \leq |N_G(W)|$$

By induction on the number $|X|$ of vertices in X .

Basis Step: $|X| = 1$. $|X| \leq |N_G(X)|$. I am married!

Induction Hypothesis: Suppose that it holds if $|X| < m$.

Induction Step: Consider the case $|X| = m$.

Consider the case $|X| = m$.

- CASE I: Every $k < m$ girls in X know $\geq k + 1$ boys in Y .

Take any girl x and marry her to any boy y she knows.

$$G' = G - \{x, y\}$$

The Hall's Condition still holds for G' .

$$\forall W \subseteq X - \{x\}. |W| \leq |N_{G'}(W)|$$

There is a $(X - \{x\})$ -perfect matching in G' .

Therefore, there is a $(X - \{x\})$ -perfect matching in G .

- CASE II: There is a set of $k < m$ girls in X who know k boys in Y .

There k girls can be married **by induction** to the k boys.

$$G' = G - \{\text{these } k \text{ girls}\} - \{\text{these } k \text{ boys}\}$$

G' satisfies the Hall's Condition.

By contradiction:

Suppose that in G' , there are $l \leq m - k$ girls who know $< l$ boys.

Then, in G , there are $k + l$ girls who know $< k + l$ boys.

There is a $(X - \{\text{these } k \text{ girls}\})$ -perfer matching in G' .

Therefore, there is a $(X - \{x\})$ -perfer matching in G .

Theorem (Hall Theorem; 1935)

Let $G = (X, Y, E)$ be a bipartite graph. There is a X -perfect matching of G *iff*

$$\forall W \subseteq X. |W| \leq |N_G(W)|$$

Definition (M -alternating Paths)

Let M be a matching. An M -alternating path is a path that alternates between edges in M and edges not in M .



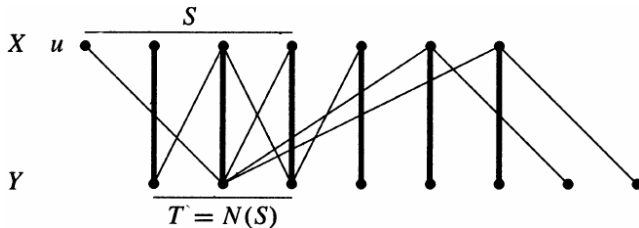
Definition (M -augmenting Paths)

An M -augmenting path is an M -alternating path whose endpoints are unsaturated by M .

By contradiction.

Suppose that there is *no* X -perfect matching.

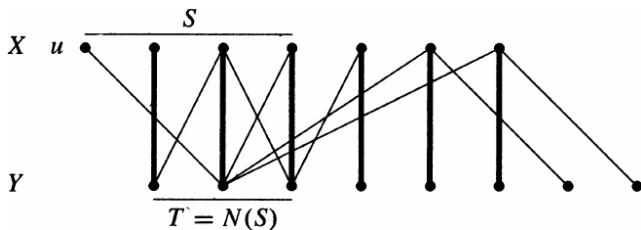
We show that Hall's Condition is violated for *some* $S \subseteq X$.



Let M be a *maximum* matching.

Let $u \in X$ be a vertex of X not saturated by M .

Consider all *M -alternating paths* starting from u .



$T \triangleq$ the set of vertices in Y reachable from u by M -alternating paths.

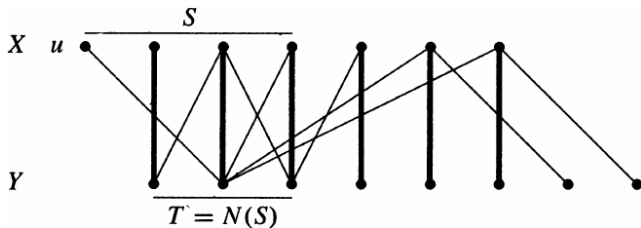
$S \triangleq$ the set of vertices in X reachable from u by M -alternating paths.

We will show that

$$T = N(S) \wedge |T| = |S - \{u\}|$$

$$|N(S)| = |T| = |S| - 1 < |S|$$

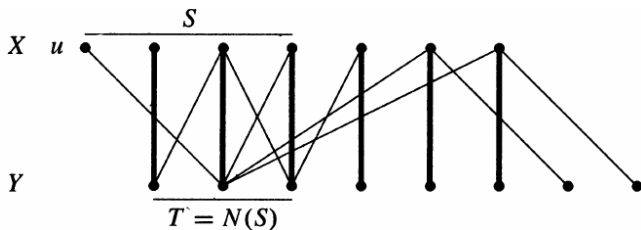
$$|T| = |S - \{u\}|$$



We show that there is a bijection from T to $S - \{u\}$.

M matches T with $S - \{u\}$.

$$T = N(S)$$



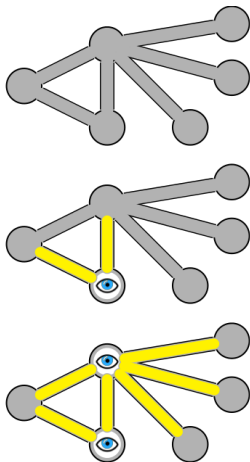
$$|T| = |S - \{u\}| \implies T \subseteq N(S)$$

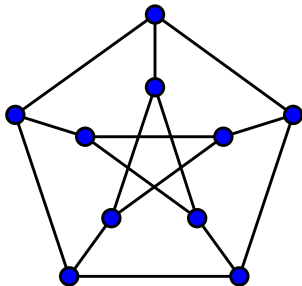
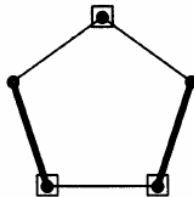
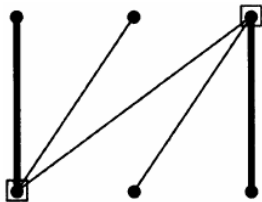
We need to show that $N(S) \subseteq T$

Consider the neighbors of $x \in S$: $x = u$, $x \in S - \{u\}$

Definitions (Vertex Cover (点覆盖))

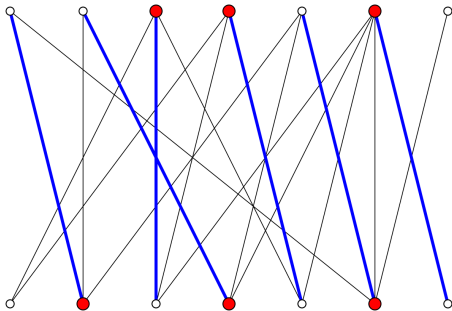
A **vertex cover** of a graph G is a set $Q \subseteq V(G)$ that **covers** all edges.





Theorem (Weak Duality Theorem (弱对偶定理))

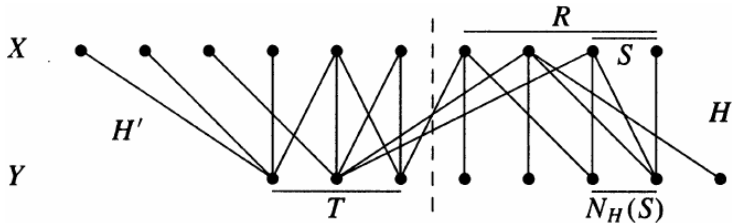
Let G be a graph. The maximum size of a matching in G
 $\leq$ the minimum size of a vertex cover of G .



Theorem (König (1931), Egerváry (1931))

Let G be a *bipartite* graph. The maximum size of a matching in G
equals the minimum size of a vertex cover of G

Given a vertex cover Q , we construct a matching M of the same size.



$$R = Q \cap X \quad T = Q \cap Y$$

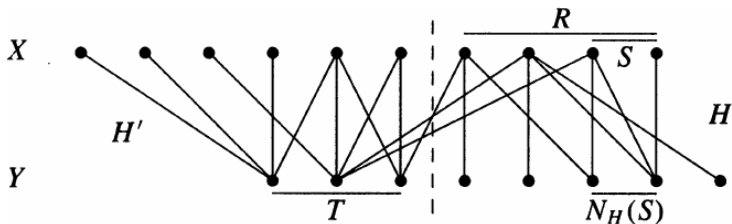
$H \triangleq (R \cup (Y - T))$ -induced subgraph of G

$H' \triangleq (T \cup (X - R))$ -induced subgraph of G

G has no edges from $X - R$ to $Y - T$.

H has a R -perfect matching and H' has a T -perfect matching.

H has a R -perfect matching and H' has a T -perfect matching.



By contradiction.

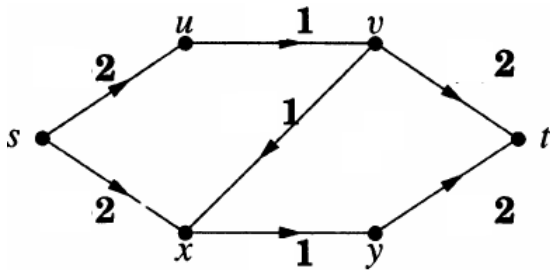
$$\exists S \subseteq R. |N_H(S)| < |S|$$

$T \cup (R - S + N_H(S))$ is a smaller vertex cover than Q

Definition (Network (网络))

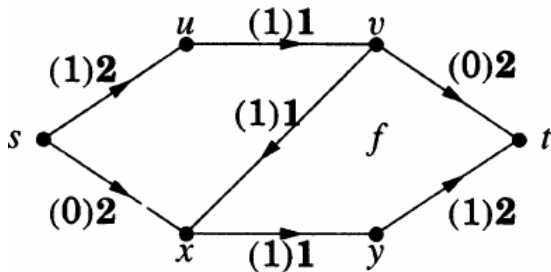
A **network** is a **digraph** with

- ▶ a distinguished **source vertex** s ,
- ▶ a distinguished **sink vertex** t ,
- ▶ a **capacity** $c(e) \geq 0$ on each edge e



Definition (Flow (流))

A **flow** f is a **function** that assigns a value $f(e)$ to each edge e .



Definition (Feasible Flow (可行流))

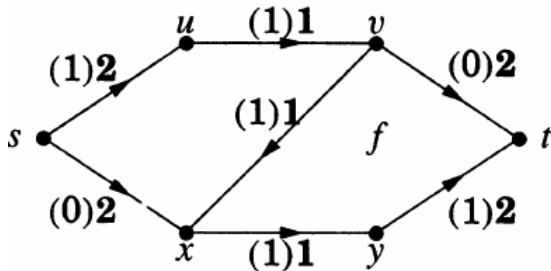
A flow f is **feasible** if it satisfies

Capacity Constraints:

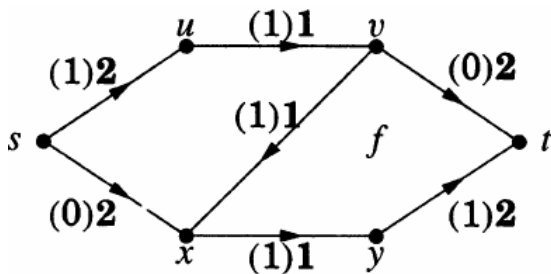
$$\forall e \in E. 0 \leq f(e) \leq c(e)$$

Flow Conservation:

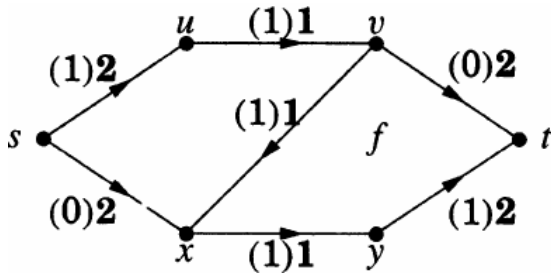
$$\forall v \in V. f^+(v) = f^-(v)$$



$$f^+(v) = \sum_{(v,w) \in E} f(v,w) \quad f^-(v) = \sum_{(u,v) \in E} f(u,v)$$



$$f^+(U) = \sum_{u \in U, v \in \bar{U}, (u,v) \in E} f(u,v) \quad f^-(U) = \sum_{v \in \bar{U}, u \in U, (v,u) \in E} f(v,u)$$



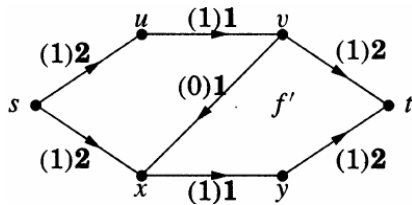
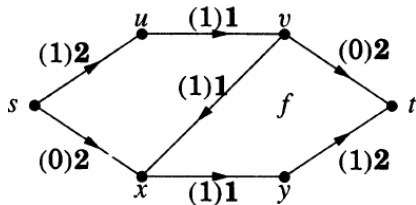
$$\forall U \subseteq (V - \{s, t\}). f^+(U) = f^-(U)$$

$$s \in U \wedge t \notin U \implies f^+(U) - f^-(U) = f^+(s)$$

Definition (Value (值))

The **value** $\text{val}(f)$ of a **flow** f is

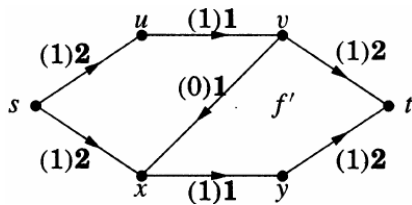
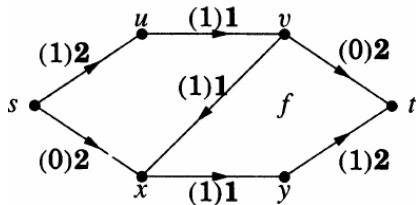
$$\text{val}(f) = f^-(t) = f^+(s).$$



Definition (Maximum Flow (最大流))

A **maximum flow** is a **feasible flow** of maximum **value**.

$$s - x - v - t$$



Definition (f -augmenting Paths (增广路径))

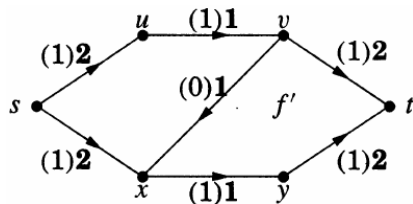
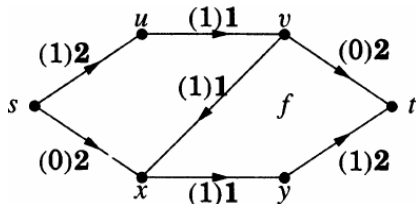
When f is a feasible flow, an **f -augmenting path** is a $s \sim t$ path P in the underlying graph such that for each edge $e \in E(P)$,

- (a) if P follows e in the forward direction, then $f(e) < c(e)$;
- (b) if P follows e in the backward direction, then $f(e) > 0$.

Definition (f -augmenting Paths)

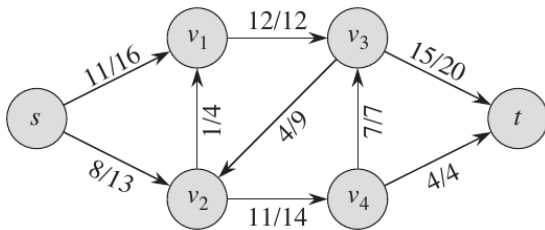
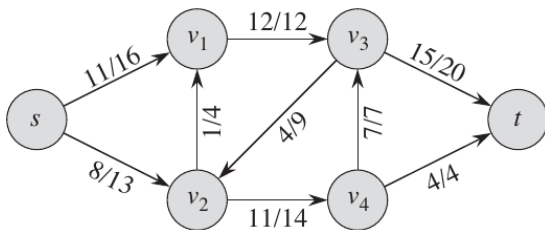
Let P be an f -augmenting path.

$$\epsilon(e) = \begin{cases} c(e) - f(e) & \text{if } e \text{ is forward on } P \\ f(e) & \text{if } e \text{ is backward on } P \end{cases}$$



An f -augmenting path leads to a flow with **larger** value.

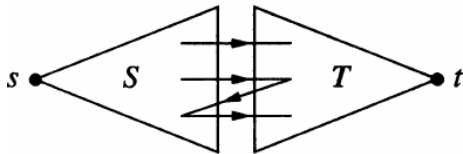
$$\min_{e \in E(P)} \epsilon(e)$$

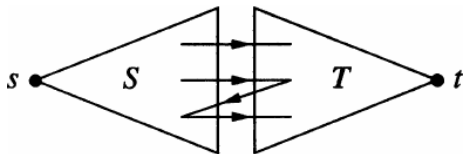


Definition (Source/Sink Cut (割))

In a network, a **source/sink cut** $[S, T]$ consists of the edges **from** a **source set** S **to** a **sink set** T , where

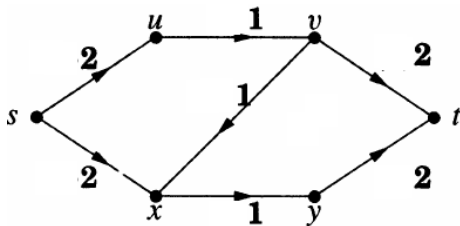
$$(T = V - S) \wedge (s \in S) \wedge (t \in T)$$





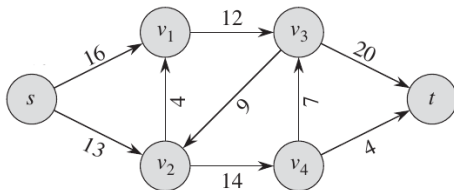
Definition (Capacity of Cut (割的容量))

$$\text{cap}(S, T) = \sum_{u \in S, v \in T, uv \in E} c(u, v)$$



Definition (Minimum Cut (最小割))

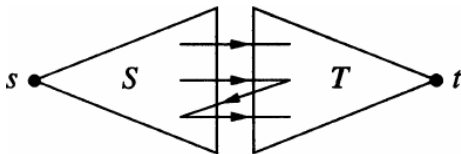
A **minimum cut** is a **cut** of minimum value.



Theorem (Weak Duality (弱对偶定理))

Let f be any feasible *flow* and $[S, T]$ be any source/sink *cut*.

$$\text{val}(f) \leq \text{cap}(S, T).$$



$$\text{val}(f) = f^+(S) - f^-(S) \leq f^+(S) \leq \text{cap}(S, T)$$

Lemma

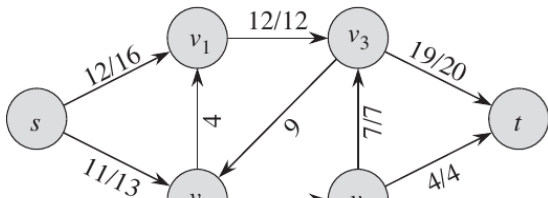
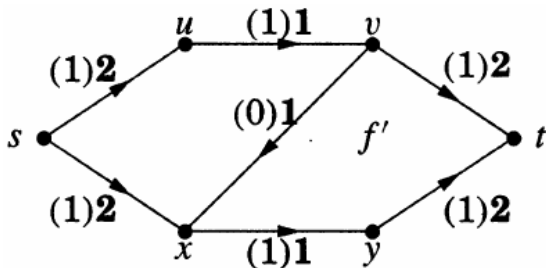
$$\max_f \text{val}(f) \leq \min_{[S,T]} \text{cap}(S, T)$$

What if $\text{val}(f) = \text{cap}(S, T)$ for some flow f and some cut $[S, T]$?

f is maximum and $[S, T]$ is minimum

$$\text{val}(f) = f^+(S) - f^-(S) = f^+(S) = \text{cap}(S, T)$$

$$f^-(S) = 0 \wedge f^+(S) = \text{cap}(S, T)$$



Theorem (Max-flow Min-cut Theorem (Ford and Fulkerson; 1956))

$$\max_f \text{val}(f) = \min_{[S,T]} \text{cap}(S, T)$$

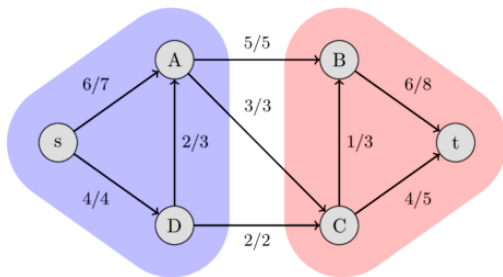
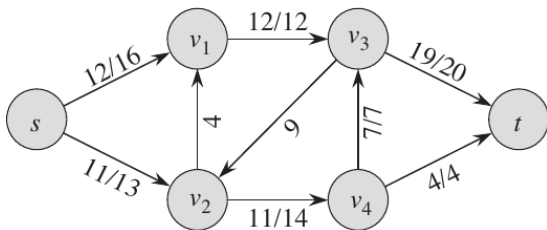
(Strong Duality)



L. R. Ford Jr. (1927 ~ 2017)



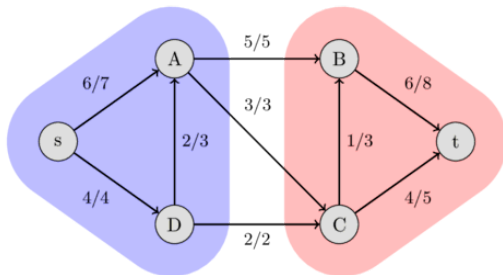
D. R. Fulkerson (1924 ~ 1976)



Theorem

A feasible flow f is maximum iff there are no f -augmenting paths.

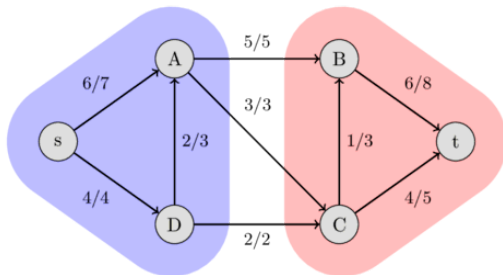
We construct a cut $[S, T]$ with $\text{val}(f) = \text{cap}(S, T)$.



$S \triangleq \{\text{the vertices reachable from } s \text{ along partial } f\text{-augmenting paths}\}$

$S \triangleq \{\text{the vertices reachable from } s \text{ along partial } f\text{-augmenting paths}\}$

$$T \triangleq V - S$$

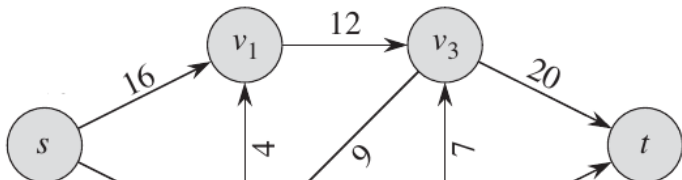
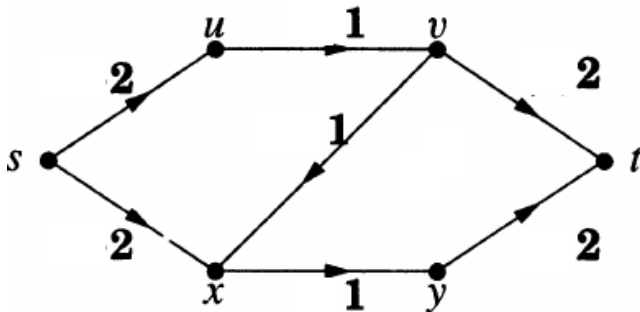


$$f^{-1}(S) = 0 \wedge f^{+}(S) = \text{cap}(S, T)$$

$$\text{val}(f) = f^{+}(S) - f^{-}(S) = f^{+}(S) = \text{cap}(S, T)$$

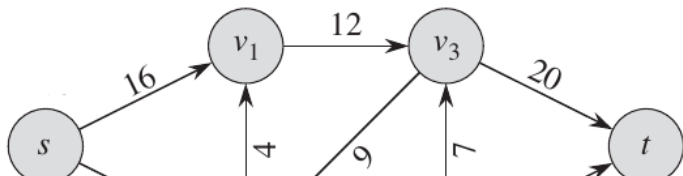
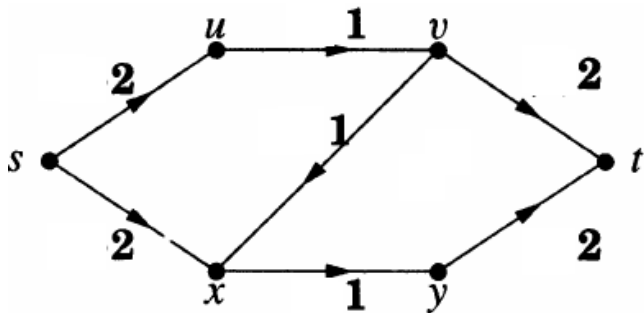
The Ford-Fulkerson Method

Repeatedly finding f -augmenting paths until no more ones exist.



The Edmonds-Karp Algorithm

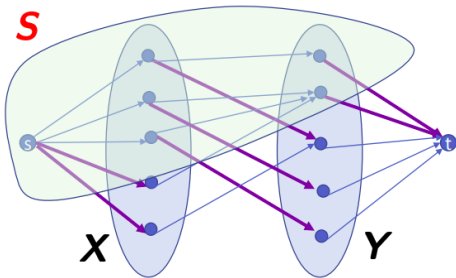
Using **BFS** (Breadth-first Search) to find f -augmenting paths.



Theorem (Hall Theorem; 1935)

There is an X -perfect matching of G iff

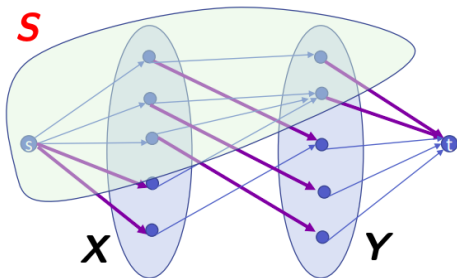
$$\forall W \subseteq X. |W| \leq |N_G(W)|$$



$$\forall x \in X. c(s, x) = 1 \quad \forall y \in Y. c(y, t) = 1 \quad \forall x \in X, y \in Y. c(x, y) = \infty$$

We need to show that $\max_f \text{val}(f) = |X|$.

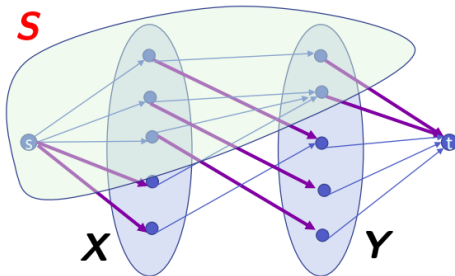
We need to show that $\min_{[S, \bar{S}]} \text{cap}(S, \bar{S}) = |X|$.



$$\min_{[S, \bar{S}]} \text{cap}(S, \bar{S}) \leq |X|$$

Therefore, we need to show that $\min_{[S, \bar{S}]} \text{cap}(S, \bar{S}) \geq |X|$.

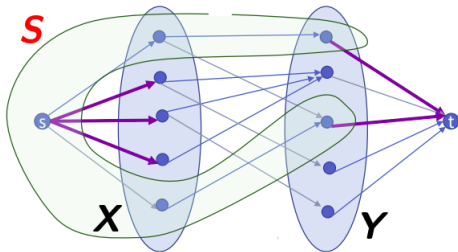
Let $[S, \bar{S}]$ be a cut. We need to show that $\text{cap}(S, \bar{S}) \geq |X|$.



$$N(S \cap X) \not\subseteq (S \cap Y)$$

$$\text{cap}(S, \bar{S}) = \infty \geq |X|$$

$$N(S \cap X) \subseteq (S \cap Y)$$



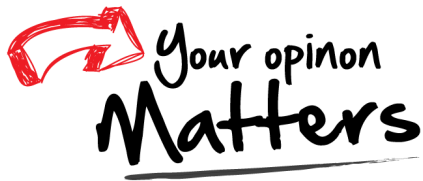
$$\begin{aligned}
 \text{cap}(S, \bar{S}) &= \sum_{u \in S, v \in \bar{S}} c(u, v) \\
 &= \sum_{v \in \bar{S} \cap X} c(s, v) + \sum_{u \in S \cap Y} c(u, t) \\
 &= |X| - |S \cap X| + |S \cap Y| \\
 &\geq |X| - |S \cap X| + |N(S \cap X)| \geq |X|
 \end{aligned}$$

Theorem (König (1931), Egerváry (1931))

If G is a bipartite graph, then the maximum size of a matching in G equals the minimum size of a vertex cover of G



Thank
You!



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